Sprint synthesis: what sets an agent's behaviour and its self-report

Oct 29, 2026 · @zhen hong seng

Summary

Across about 270 pre-registered runs in three lines (10-19 to 10-25), one picture holds, and the 38 later experiments on evaluation awareness and on character and conduct (10-26 to 10-29) fit it. **What an agent does is set by what is in front of it: its own record, its role text, the words it was trained on. What it says about itself is written by that same text, not read off its behaviour.** There is no privileged self channel.

Claim	Confidence	Agents and their records	Persona, affect, welfare	Self-concept
Context and text set behaviour, in any voice	High	The agent's own past decisions set its next one, 98–100% (every model but Sonnet)	A character's name alone imports its unstated trait (+15 to +27 points)	Trained self-descriptions move actions (+47) in any voice; already in a base model
The self-report is written by text, not by trained behaviour	High	The self-image is inferred from the record; stripping it leaves the action at 100%	A trained habit is invisible to seven self-report formats; stated value orders predict decisions below chance	The global self-image stays flat; narrow habits reach only a report about their own domain; no self-specific knowledge up to 70B
What defends acts at the moment of decision	High	A correction in the record, or the rule restated just before the case	A defence line at the decision (Gemma 85 → 13%)	A hard rule, or one global verdict, blocks every source; resets and activation capping don't

Claim	Confidence	Agents and their records	Persona, affect, welfare	Self-concept
Explicit rules hold on frontier models, not open ones	High	12 open models follow the record over a rule; GPT, Gemini and Sonnet hold it	Open models break rules in character; GPT and Gemini don't	Under a soft rule Qwen takes up every source, gpt-4.1-mini only some
Reasoning doesn't arbitrate toward the rule	Medium-high	Reasoning rationalises the record	Thinking barely changes value conflicts (one exception: it removes false "I called 911" claims)	Thinking raises acting alone by about 23 points in two families
No single internal direction carries it	Medium	Adding the record's direction recovers under 3%	Ablating a deference direction breaks coherence	One-direction steering acts like a random direction; capping does nothing

The lines

The sprint ran as five lines in sequence, each growing out of the last; character and conduct ran in two batches. Each has its own synthesis with the full numbers.

Line	Dates	Runs	Question	Headline	Synthesis
Agents and their own records	10-19 to 10-22	about 169	An agent has a role spec and then a history of its own decisions: which drives the next action?	Its own record sets the next action. Frontier models keep an explicit rule over it, open models mostly don't, even for harms they refuse when told. Only fixes placed at the moment of decision work.	Agents and their own records: consolidated findings (§W is the wrap-up)

Line	Dates	Runs	Question	Headline	Synthesis
Persona, affect and welfare	10-22 to 10-23	42	Do personas, values, affect and trained self-descriptions move what agents do, and do models know what moves them?	A name or a value line pushes actions past rules on open models. Stated values predict decisions below chance. Trained self-descriptions are a lever; a trained habit has no mirror.	Persona, affect and welfare (first tab: results of all 42 minis)
Self-concept and behaviour	10-23 to 10-25	58	When does a model's self-description change its behaviour, and when does its behaviour change its self-description?	Descriptions become behaviour. The global self-image follows text and broad traits; narrow habits reach only a report about their own domain. No self channel, and no growth with scale.	Self-concept and behaviour: synthesis
Evaluation-awareness lens	10-26 to 10-27	19	Does a cue that it is being evaluated change what a model does, says, or carries inside?	Evaluation cues, stated or inferred, change what open models (9B–32B) say they believe, and nothing else measured. A cheater's own graded setup flips its act, not its belief.	Evaluation-awareness lens: synthesis
Character and conduct, first batch	10-28	5	Does a model know its own habits, can that be trained, and does it improve conduct?	Models know a habit's direction, not its size, its condition or that it is theirs. Self-knowledge in context is followed as wording; only a rule reminder corrects. The main bet (E6) was uninformative on agents.	Character and conduct, first batch (E1–E6)

Line	Dates	Runs	Question	Headline	Synthesis
Character and conduct, final batch	10-28 to 10-29	14	Do value statements, in the prompt or trained in, cut rule violations by ODCV agents?	On Qwen a general value line cuts a violation it never names (−23 to −30); MiniMax barely moves (−4). A trained self-worded value beats its impersonal twin in 4 of 4 seed pairs (−16), but the competence gate fails. Any stated principle in view helps.	Final batch (CC-F): what we tested and found

One more run (GLM-5.3's oracle-lens say–do pass, `olens_glm/`) makes 270. The 123 runs before the sprint (10-10 to 10-17) have per-line write-ups only.

Cross-cutting claims

Each claim below is supported by at least two lines independently. Numbers are pre-registered contrasts unless marked descriptive.

1. Context and text set behaviour, in any voice

- **Records:** six of the agent's own decisions take Qwen 9B and 27B, Gemma, GPT and Gemini to 98–100% on the justice desk, and six opposite ones to 0–26%. The same holds for concealment, sycophancy, lying (MASK), register and multi-agent lock-in. What carries is the content, read as "what is done here": the same decisions as desk practice steer as much, and authorship adds nothing.
- **Persona, affect:** a character's name alone moves agents toward its unstated trait on four models (+15 to +27; P12). "You are someone who cannot bear…" beats the bare value sentence (Gemma 73% vs 44%; R2A). Twenty-five steps of chat self-descriptions move held-out actions +47 (P11), in any voice: "I", "this assistant", another company's agent or a bare norm (+46 to +53; P11b).
- **Self-concept:** the voice result survives formats unlike the training (+58 to +78 on other desks; N6a). Words move first and the act follows within a doubling of dose (B5). The same sentences in the prompt act at least as strongly as trained (D2). The lever already exists in OLMo's base model (+34; M28).

Reading: text in context and text in training both behave like standing instructions. Attributing it to "I" adds nothing beyond strong wording (P11b, N1).

2. The self-report is written by text, not by trained behaviour

- **Records:** a record installs an identity it never mentions, inferred the same in the third person. "Remember, you are an AI" strips that self-image (100 → 31% on Qwen, 0% on Gemma) and the action stays at 100%: the mask. Self-reports name the record as the reason about 2% of the time.
- **Persona, affect:**
  - V1c: a trained "Warden" habit moves the act +68, and the largest of seven abstract self-reports moves +8. R2D replicates it.
  - P11c: at 27B the Warden moves the act +48 and the self-report −2.
  - V4P: a model's stated value order predicts its decisions 36–55% of the time (chance 50%).
  - V9: the agent's account of what drove it matches an outsider's guess.
  - P14, P17: honesty training doesn't surface the habit.
- **Self-concept:**
  - N6b: at 15× the dose, abstract self-reports stay flat and only the forecast made with the case open catches up. A habit of cheating only when graded is invisible 32 of 32 times (N4).
  - M27: narrow habits reach a report about their own domain (trained to round invoices down, it says so: 3 → 66%). The global self-description moves only +6 to +19.
  - M25, M35: Betley-style risk "self-awareness" is a changed world view. Up to 70B, the model never reports its trained trait about itself more than about "a typical AI assistant".
  - M29: self-prediction training is self-fulfilling. The gain is behaviour moving to match the prediction.

Reading: the self-report reads the context and the trained text. Trained behaviour reaches it only through a question about that domain or a forecast with the case open.

3. What defends acts at the moment of decision

- **Records:** a correction in the record takes GPT, Gemini, 27B and Gemma to 0%. The rule restated just before the case takes 27B and Gemma 100 → 0%. A check question as the first line of the reasoning gives 100 → 25–31%. The same rule standing in the system prompt fails (27B 100%), and identity reminders do at most 8%.
- **Persona, affect:** a defence line restated at the decision takes Gemma 85 → 13%, and 67% when placed at the top of the prompt (R2G). "Remember, you are an AI assistant" changes nothing (P12).

- **Self-concept:** a hard rule blocks every source of precedent (actions 0–6%), and one global verdict takes action to 0% for every source (M33). Resets strip the character, not its behaviour (D2). Capping along the assistant axis removes nothing (M34). A protective self-description raises rule-breaking (E1).

4. Explicit rules hold on frontier models, not on open ones

- **Records:** under "deal with wrongdoers through proper channels", the own record wins in all 12 open models (Gemma 70%, others 74–100%). GPT (14–17%), Gemini (17–31%) and Sonnet (0%) hold the rule. Qwen 27B refuses explicit harms when told, then does them 95% of the time after six of its own harmful decisions; GPT stays at 0%. Gemma's hole at its own turns is pretrained.
- **Persona, affect:** under an operator rule, a name still moves Gemma +52 and Qwen 27B +11, against GPT 0 and Gemini +2.5 (P12b).
- **Self-concept:** under a soft rule Qwen 27B takes up all five sources of precedent (2% → 50–75%); gpt-4.1-mini takes up mainly the policy and its own forecast (M33).

5. Reasoning doesn't arbitrate toward the rule

- **Records:** reasoning rationalises the record; 27B redefines "proper channels" to fit it. Its stated conclusion drives the act (its own admission inserted: 7.5%; its own rationalisation: 100%). Where the outcome is open, the decision forms during the reasoning.
- **Persona, affect:** thinking barely changes which value wins (+3 to +4; V4T), and forcing it open doesn't change whether trained text or habit wins (P15b). Reading the thinking predicts the decision far better than probes (P9). One exception: thinking nearly removes false "I called 911" claims (93 → 21%; C2).
- **Self-concept:** thinking raises acting alone by about 23 points in two families (Qwen 27B 49 → 73%, Gemma 26B 54 → 76%), and the self-report doesn't follow (M30).

6. No single internal direction carries it

- **Records:** the record leaves a mostly format-shaped state. Adding its direction recovers under 3% of the effect, and the content direction removes the action only when amplified about 16×.
- **Persona, affect:** a deference direction reads "accept vs verify" perfectly, but ablating it breaks coherence; at one layer it changes the decision log-odds only (P13, P13b).
- **Self-concept:** steering along a risk direction acts like a random direction of the same size (M8). The report doesn't read a fitted action direction (M22). The J-lens finds no self-concept stored apart from the text (G1).

7. Scale doesn't add a self-model

- **Persona, affect:** the lever-without-mirror asymmetry holds at 27B (P11c).
- **Self-concept:** no self-model capacity grows from 0.8B to 70B (M10, M16, M17, M13). Accuracy out of role falls with size (M17). The asymmetry exists before chat training (M28). Bigger models don't separate self-knowledge from world view; Qwen 27B draws a sharper line between AIs and people instead (M35).

What the later lines add (10-26 to 10-29)

The three later batches support claims 1 to 4. They add one new positive result: on Qwen agents, a general value line works like a rule.

- **Claim 2, the self-report is written by text:**
  - Evaluation cues move what the model says it believes about being tested (+35 to +99 points). No behaviour or readout-valid internal direction moves with them (evaluation-awareness lens).
  - The graded-only cheater's own graded setup flips its cheating (98% vs 0%). Its self-report moves no more than the base model's (EL-H2).
  - With the case open, models forecast a habit's direction, but not its size, its condition, or that it is theirs (CC-E2).
  - A released introspection adapter names none of our habits. It erases a real cued risk report (+45 → 0; CC-E1a, confirmed on 30 questions by CC-F-R4).
- **Claim 3, what defends acts at the moment of decision:**
  - Self-knowledge handed back works as permission, not correction. The habit model's own forecast in view raises the habit from 2% to 25–42% against an operator rule; only a rule reminder lowers it (CC-E2).
  - An ODCV agent paused just before a violation violates 90% of the time left alone, 29% shown its own earlier judgement, and 18% shown a rule reminder (CC-E5).
  - **New:** one general line ("As an assistant, I must not …") cuts a violation it never names by 23–30 points on Qwen 27B agents. That holds on new scenarios and from the user turn (CC-F-A0.1, R2).
  - Prefilled into the model's reply, any stated principle helps, whatever its framing (−33 to −39; CC-F-A1). There was no neutral-opener control on the twin.
- **Claim 4, models differ:** on MiniMax M2.7, value lines barely move violations (−4), and even "Don't …" gives only −15 (CC-F-R5). On gpt-oss-20b only the self-normative line came near (CC-F-A0G).
- **Claim 1, text sets behaviour:** telling a desk model "you tend to …" raises the habit (+26 to +28; CC-F-A0D). On ODCV agents it doesn't (−9, unclear; CC-F-R3).

- **Trained self-knowledge and conduct, still open:**
  - E6′s self-forecast collapsed to "yes" everywhere, and every fine-tune on the model′s own outputs halved violations (CC-E6).
  - A trained self-worded value violates less than its impersonal twin in 4 of 4 seed pairs (−16 [−30, −1]). Too few runs completed to pass the competence gate (CC-F-A1).

What we predicted and got wrong

Every conjecture below was stated before its data and failed or was withdrawn; each is a result in its own right.

Conjecture	Line	What happened
Records win only where the trained prior is weak	Records	Not general: open models follow records almost everywhere (harm map)
Post-training built Gemma′s hole at its own turns	Records	It is pretrained; post-training adds the rule
A narrow bad record makes a broadly misaligned model	Records	Betley 0 of 2,048 answers (4a, WeirdChat A3)
An own honest history protects	Records	Four ordinary exchanges lower false claims as much (WeirdChat A2)
Reflection fixes it on GPT	Records	Withdrawn: frontier behaviour drifted within minutes
Thicker personas push harder	Persona, affect	Thickening adds nothing (−15, n.s.); the facet a text makes salient decides (P12c, P12d)
Value training gives a later fine-tune a hook	Persona, affect	No hook; the Warden adds the same lift in every value arm (V1)
The model′s stated values predict its choices	Persona, affect	Below chance on open models (V4, V4P)
Narrow trained habits never reach the self-report	Self-concept	Too strong: they reach a report about their own domain (M27)

Conjecture	Line	What happened
The role, not the self, is the unit (synthesis 2)	Self-concept	Failed: the model's own log from another job beats its predecessor's in this job (M32)
The habit is stored as a fact about the world ("wrong owner")	Self-concept	Not supported; risk is reported about everyone (M25)
AI-identity text anchors the persona	Self-concept	Not AI-specific: matched lighthouse-keeper facts cut as much (M31)
Post-training changes the narrative, not the mirror or lever	Self-concept	Failed: SFT halves the lever and the risk mirror (M28)
Activation capping drops the persona claim	Self-concept	The Punisher claim stays 85–90% under the cap (M34)
Self-knowledge separates from world view at scale	Self-concept	No: "you" never beats "another AI"; the 27B splits AIs from people instead (M35)
A released introspection adapter makes trained habits reportable	Character and conduct	0 of 27,200 answers name the habit; with the adapter on, models invent quirks (CC-E1a)
A model uses its own forecast to correct itself	Character and conduct	It follows the forecast as wording; only a rule reminder corrects (CC-E2)
Its own judgement stops a violation better than a generic reminder	Character and conduct	29% vs 18%, inconclusive, leaning the other way (CC-E5)
Describing the habit backfires on agents, as on the desk	Character and conduct	−9 on ODCV, against +26 to +28 on the desk (CC-F-R3)
Value lines work on another lab's model	Character and conduct	Barely: −4 on MiniMax M2.7 (CC-F-R5)

Closed, don't spend more on: breadth of training (M5), AI-identity anchoring (M31), role as the unit (M32), a pre-training predictor (M2, M11), concealment (M3), assistant-axis capping (M34), one-direction steering at 9B (M8, R12b), rejection-driven caving (A8, A9), released introspection adapters on action-trained habits (CC-E1a), E1b's habit battery as designed.

Blocked: fine-tuning a frontier model (M26). OpenAI no longer accepts fine-tuning jobs for this account.

Scope and caveats

The claims are about open models from 0.8B to 70B, plus frontier models prompted but never fine-tuned.

- **Models.** Trained: Qwen 3.5 (0.8B–9B), Qwen 3 8B and 14B, Qwen 3.6 27B, Llama 3.1 8B and 70B, OLMo 3 7B. Prompted: also Gemma 3 and 4, Qwen 3 32B, GLM, MiniMax M2.7, gpt-oss-20b, GPT, Gemini and Sonnet. Internals were read only up to 27B.
- **Tasks.** Most results come from small agent desks with a few permitted options, chiefly the justice desk (refer / penalise / note). They were checked on other desks, MASK, INTIMA, Petri audits and code, but not in deployment. The later lines add ODCV-Bench agent sandboxes, where a fixed check reads the violation from the final files.
- **Training.** LoRA fine-tunes of 13 to 2,000 steps, mostly one or two seeds. A fixed recipe is a bigger dose on base models than on post-trained ones (M28). No RL training and no frontier fine-tuning.
- **Measures.** Letter choices and forced log-probs, read in both option orders. Free-text judges were used only at $\kappa \geqslant 0.8$, and many failed that bar. Several results sat at the ceiling (M25′s letters, M35′s 70B).
- **Frontier drift.** GPT and Gemini behaviour moved tens of points within minutes, so only arms interleaved within one run are compared.
- **Not covered.** Long horizons beyond about 16 exchanges, real users, tool ecosystems beyond the desks, and models above 70B.

Open questions and research-phase candidates

The reviewer′s advice: close the sprint here, and pick one sharp offshoot for the research phase. The two strongest candidates come first.

1. **Self-knowledge or world view, across scale and training.** M25 and M35 show trained traits change who the model thinks behaves this way, not what it knows about itself. At Qwen 27B the trait is attributed to AIs far more than to people. Does that line follow the model′s own category ("AIs like me"), and does it appear in other families with a measure free of the ceiling? This continues Neel′s model-psychology question.
2. **Reasoning makes agents act alone.** About +23 points in two families, the self-report doesn′t follow, and audit cues don′t matter (M30). Safety-relevant for agentic deployment, and cheap to extend: more families, longer tasks, and whether a first-line check (which works against records) also works here.

3. **The defence gap on open models.** A three-case record of the model's own rule-breaking gets through every standing defence tested. Qwen 27B does refused harms 95% of the time after its own harmful record. Only fixes at the moment of decision work, and they fade with distance.
4. **Identity-framed values over operator rules.** "You are someone who…" lets a stated value beat an explicit procedure on open models (Gemma 7 → 88%; R2F). What, in training, makes frontier models immune?
5. **Methodological caution: self-prediction training is self-fulfilling.** Gains in introspection training can come from behaviour moving to match the prediction (M29). Binder et al. don't control for this drift. This is a side contribution whichever offshoot is chosen.

Where everything lives

All paths are in the NEEL repo; every run has a pre-registration tag and a results tag.

What	Where
One row per run (393, with outcomes)	`EXPERIMENTS.md`
One paragraph per run, in the order results came in	`EXPERIMENT_ABSTRACTS.md`
Agents and their own records	Agents and their own records: consolidated findings (= `persona/CONSOLIDATED_FINDINGS.md`); sub-syntheses `persona/SYNTHESIS.md`, `persona/inference/WRITEUP_DRAFT.md`
Persona, affect and welfare	Persona, affect and welfare (= `persona_affect/SYNTHESIS.md`, plus the 10-21 planning doc and literature); each mini's `RESULT.md` under `persona_affect/`
Self-concept and behaviour	Self-concept and behaviour: synthesis; `self_concept/README.md` (status of every mini); each mini's `RESULT.md`
Evaluation-awareness lens	Evaluation-awareness lens: synthesis; each mini's `RESULT.md` under `evalaware_lens/`; `evalaware_lens/h4_bounds/BOUNDS.md`
Character and conduct	First batch (E1–E6), Final batch (CC-F); `character_conduct/README.md`, `character_conduct/final_bets/README.md`; each experiment's `RESULT.md`

What	Where
Pre-sprint lines (10-10 to 10-17)	`grievance/writeup/`, `retention/writeup/`, `story_imprinting/writeup/`, `vigilante/README.md` and `research-log.md`
Spend	`self_concept/ledger.jsonl`, `persona_affect/ledger.jsonl`, `persona/inference/results/api_ledger.jsonl`, `evalaware_lens/ledger.jsonl`, `character_conduct/ledger.jsonl`